Wine Quality Analysis

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Problem overview

- Goal
  - Predict the wine quality based on physicochemical properties from the datasets
  - Classify wines into "good" (quality > 6) and "bad" (quality ≤ 6) categories.
- Steps
  1. Create binary target variable
  2. Split data: Create training, validation, and testing sets
  3. Train a classification model: Using at least 3 algorithms
  4. Adjust hyperparameters using validation set
  5. Evaluate the model

Dataset Description

- Source: UCI Machine Learning Repository
  - Collected from Portuguese wines
  - Based on physicochemical laboratory tests
- Characteristics
  - 4898 samples
  - 11 features
  - 1 output variable(Quality score)
  - No missing values
  - Red wine and white wine combined for analysis

Variables

Variable Name	Role	Type	Description	Units	Missing Values
fixed_acidity	Feature	Continuous			no
volatile_acidity	Feature	Continuous			no
citric_acid	Feature	Continuous			no
residual_sugar	Feature	Continuous			no
chlorides	Feature	Continuous			no
free_sulfur_dioxide	Feature	Continuous			no
total_sulfur_dioxide	Feature	Continuous			no
density	Feature	Continuous			no
pH	Feature	Continuous			no
sulphates	Feature	Continuous			no
alcohol	Feature	Continuous			no
quality	Target	Integer	score between 0 and 10		no
color	Other	Categorical	red or white		no

Data Preparation

1. Combined datasets
2. Created binary target variable
3. Good wine is quality > 6 | Bad wine is quality ≤ 6
4. Data split into 3 sets: training(70%), validation(15%), testing(15%)
5. Applied feature scaling using StandardScalar

Classification Model Methods

- Decision Tree
  - Rule-based classification: The algorithm splits the dataset into branches using feature-based decision rules until it reaches a final classification
  - Hierarchical structure: Data moves from the root node through a series of decisions (splits) that aim to separate classes as clearly as possible
  - Hyperparameter control: Model complexity is mainly controlled by parameters like max_depth, which prevents the tree from growing too large and overfitting the data.
- Nearest Neighbors
  - Distance-based classification: A new data point is classified based on the labels of the K closest training samples
  - Similarity assumption: The algorithm assumes that data points with similar feature values are likely to belong to the same class
  - Key parameter (K): The number of neighbors determines how predictions are made; small values of K are sensitive to noise, while larger values produce smoother predictions

Classification Model Methods

- Naïve Bayes Classifier
  - Probability-based model: Uses Bayes' Theorem to compute the probability that a data point belongs to a particular class.
  - Feature independence assumption: Assumes that all input features contribute independently to the probability of a class label.

Model Evaluation Metrics

- Accuracy
  - Percentage of correct predictions
- Precision
  - How many predicted good whines were actually good
- Recall
  - Ability to detect good whines
- F1 score
  - Balance between precision and recall

Results

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Validation Results
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Decision Tree
Accuracy : 0.8430769230769231
Precision: 0.5989847715736041
Recall   : 0.6145833333333334
F1 Score : 0.6066838046272494
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KNN
Accuracy : 0.837948717948718
Precision: 0.5977011494252874
Recall   : 0.5416666666666666
F1 Score : 0.5683060109289617
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Naive Bayes
Accuracy : 0.7569230769230769
Precision: 0.4204946996466431
Recall   : 0.6197916666666666
F1 Score : 0.5010526315789474
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Test Results
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Decision Tree
Accuracy : 0.841025641025641
Precision: 0.5957446808510638
Recall   : 0.5863874345549738
F1 Score : 0.5910290237467019
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KNN
Accuracy : 0.8369230769230769
Precision: 0.6012658227848101
Recall   : 0.4973821989528796
F1 Score : 0.5444126074498568
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Naive Bayes
Accuracy : 0.7589743589743589
Precision: 0.42028985507246375
Recall   : 0.6073298429319371
F1 Score : 0.4967880085653104
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Analysis

- Decision Tree
  - Best performing due to overall balance between results
  - Highest F1 score
- KNN
  - Similar accuracy to decision tree
  - Lower recall
- Naive Bayes
  - Weak precision
  - Predicts too many wines as good

Future work

- Implementation of additional machine learning algorithms
- Use a more balanced dataset
- Use the variables that would be more specific to the goal
- Optimize the tuning to perform better by having better parameters
- Have code that detects outliers too the dataset
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https://github.com/nlewicz/LAweek9Q3.git